FLORIDA ATLANTIC UNIVERSITY

CAP 6776 · Information Retrieval · Course Project

Biomedical Hybrid Information Retrieval



Combining lexical relevance, semantic understanding and cross-encoder reranking on NFCorpus

PRESENTER

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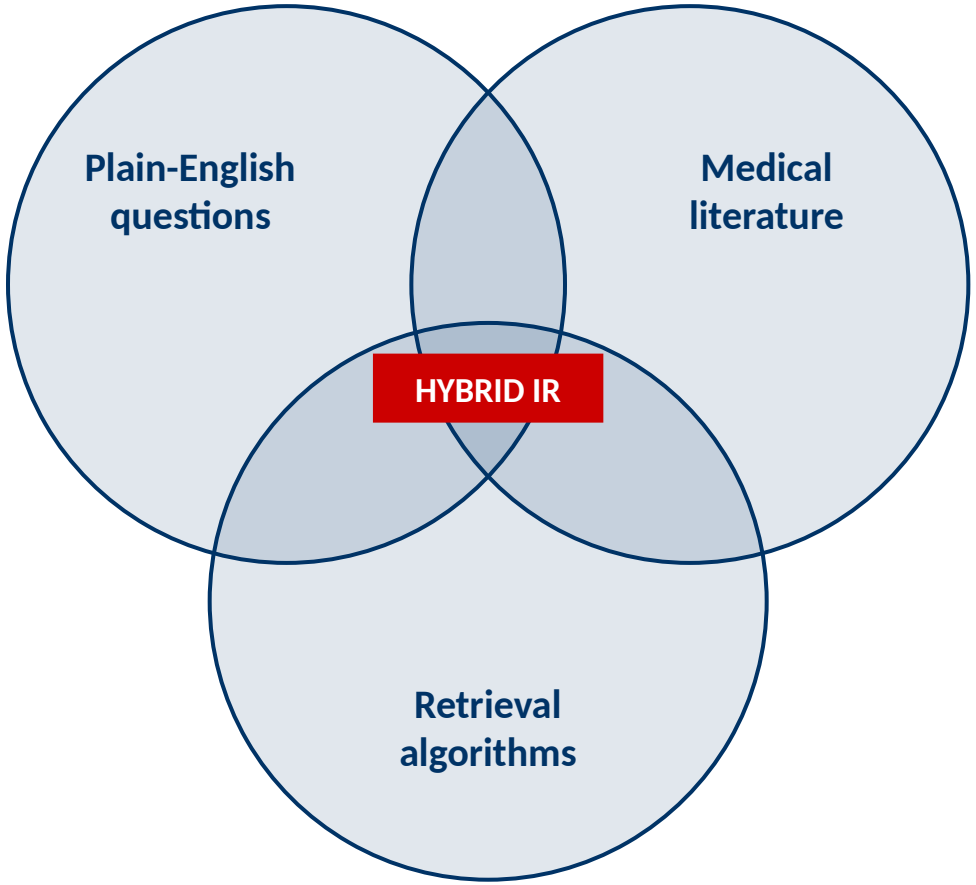
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COURSE

CAP 6776 · Information Retrieval

PROJECT OVERVIEW

Bridging everyday health questions and scientific evidence



The mission

Design a hybrid biomedical retrieval system that ranks PubMed-style articles for natural-language health questions.

Core components

1. **BM25** — Lexical keyword matching
2. **MedCPT** — Dense semantic retrieval
3. **RRF** — Reciprocal Rank Fusion
4. **Cross-encoder** — Final reranking of the top 50

THE PROBLEM

People search in plain words; papers use medical vocabulary

USER TYPES

“canker sores”



RELEVANT PAPER SAYS

“…etiology of recurrent aphthous ulcers”

Result: Zero shared words — keyword search cannot find the paper.

USER TYPES

“Fosamax”



RELEVANT PAPER SAYS

“Bisphosphonate-associated osteonecrosis of the jaw”

Result: BM25 returns an empty list for this query.

Real NFCorpus test queries (PLAIN-817 and PLAIN-1214) with their official relevance judgments.

WHY IT MATTERS

The high stakes of biomedical search

Question

A non-expert asks a serious health question in plain language.

Gatekeeper

Strict lexical matching discards documents that lack the exact terminology.

Hidden evidence

Relevant clinical evidence stays hidden from the person who needs it.

The goal: combine exact keyword matching with meaning-based semantic matching — and measure the impact of this hybrid approach honestly.

EXISTING APPROACHES

Each classical approach solves only half the problem

Terminology precision ↑

Classical lexical (TF-IDF, BM25)

Strength: pinpoints exact scientific terms.

Weakness: fails on synonyms and lay wording.

The gap: hybrid target

Precise terms and broad meaning together

Neither precise nor semantic

Dense semantic retrieval

Strength: captures the broad intent of a question.

Weakness: semantic drift — e.g. “salmon” matched to generic seafood.

Semantic understanding →

RESEARCH QUESTIONS

Three questions this project sets out to answer

1 Lexical vs. dense

RQ1

Does dense semantic retrieval significantly outperform classical lexical methods (BM25) in overcoming vocabulary mismatch?

2 General vs. domain-specific

RQ2

Does domain-specific contrastive training (MedCPT) yield measurable gains over a general-purpose dense model (BGE)?

3 The cost of intelligence

RQ3

Do Reciprocal Rank Fusion and cross-encoder reranking justify their extra computational latency?

DATASET & SETUP

BEIR / NFCorpus: lay questions, expert-graded relevance

3,633

Documents (PubMed titles + abstracts)

3,237

Lay-language queries

323

Held-out test queries — the only source of reported metrics

0 / 1 / 2

Graded relevance labels, not binary

Zero-shot integrity rule

- No tuning on the test split.
- BM25 uses textbook defaults: k1 = 1.2, b = 0.75.
- RRF uses the standard k = 60.
- Every component evaluated on unseen data.

RETRIEVERS COMPARED

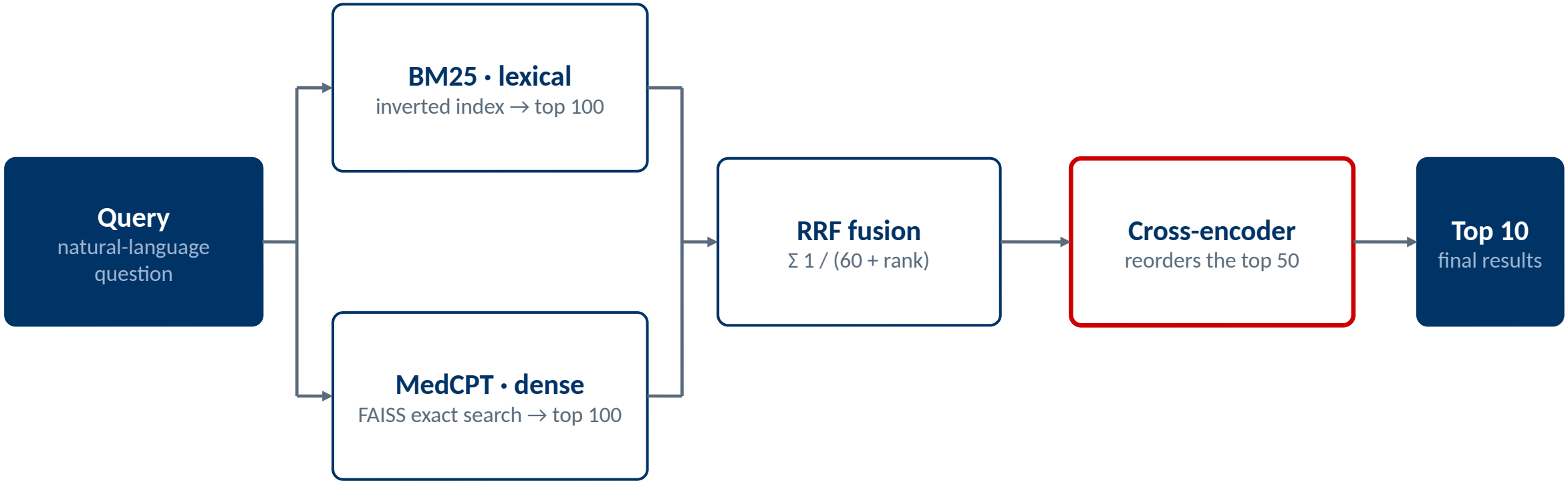
Parallel retriever baselines

Family	Matches on	Index	Core configuration
Lexical (TF-IDF / BM25)	Words / tokens	Inverted index (NFKC, lowercase, punctuation stripped)	k1 = 1.2 (term saturation), b = 0.75 (length normalization)
Dense general (BGE)	General meaning	768-d vectors, FAISS exact inner product	BAAI/bge-base-en-v1.5
Dense biomedical (MedCPT)	Clinical meaning	768-d vectors, FAISS IndexFlatIP (exact)	Separate query / article encoders trained contrastively on PubMed search logs

All three families retrieve their top 100 candidates independently for every query.

SYSTEM ARCHITECTURE

Two retrievers, fused by rank, then re-read by a cross-encoder



Mean latency per query: BM25 2.2 ms · MedCPT 1.9 ms · Hybrid RRF 4.1 ms · + reranking ≈ 1,946 ms (Apple M1 Pro)

ADVANCED TECHNIQUE

Reciprocal Rank Fusion: combine ranks, not scores

The problem



Raw scores live on incompatible scales. Adding them would let MedCPT dominate the fused list entirely.

The solution

$$\mathrm{RRF}(d) = \sum_i 1 / (60 + \mathrm{rank}_i(d))$$

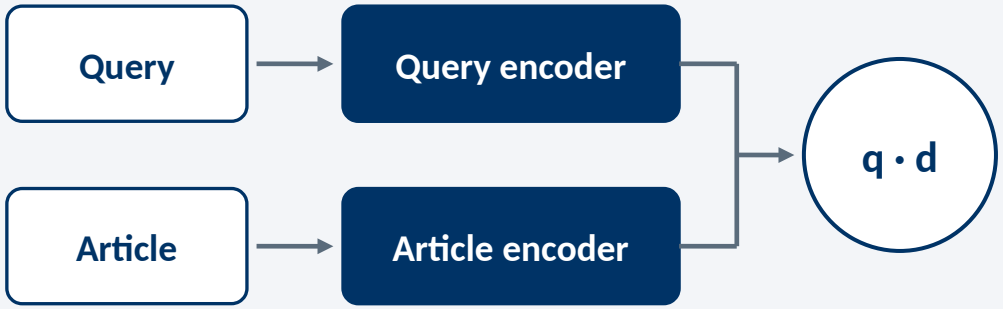
Document	BM25 rank	MedCPT rank	RRF score
Found by both	3	5	**0.0313**
MedCPT only	—	1	0.0164

Agreement between both retrievers outranks a single first place. k = 60 keeps one list from dominating.

FINAL STAGE

Cross-encoder reranking: more accurate, far slower

Bi-encoder (retrieval)



Encoded separately; articles indexed offline. Fast vector lookup.

Cross-encoder (reranking)

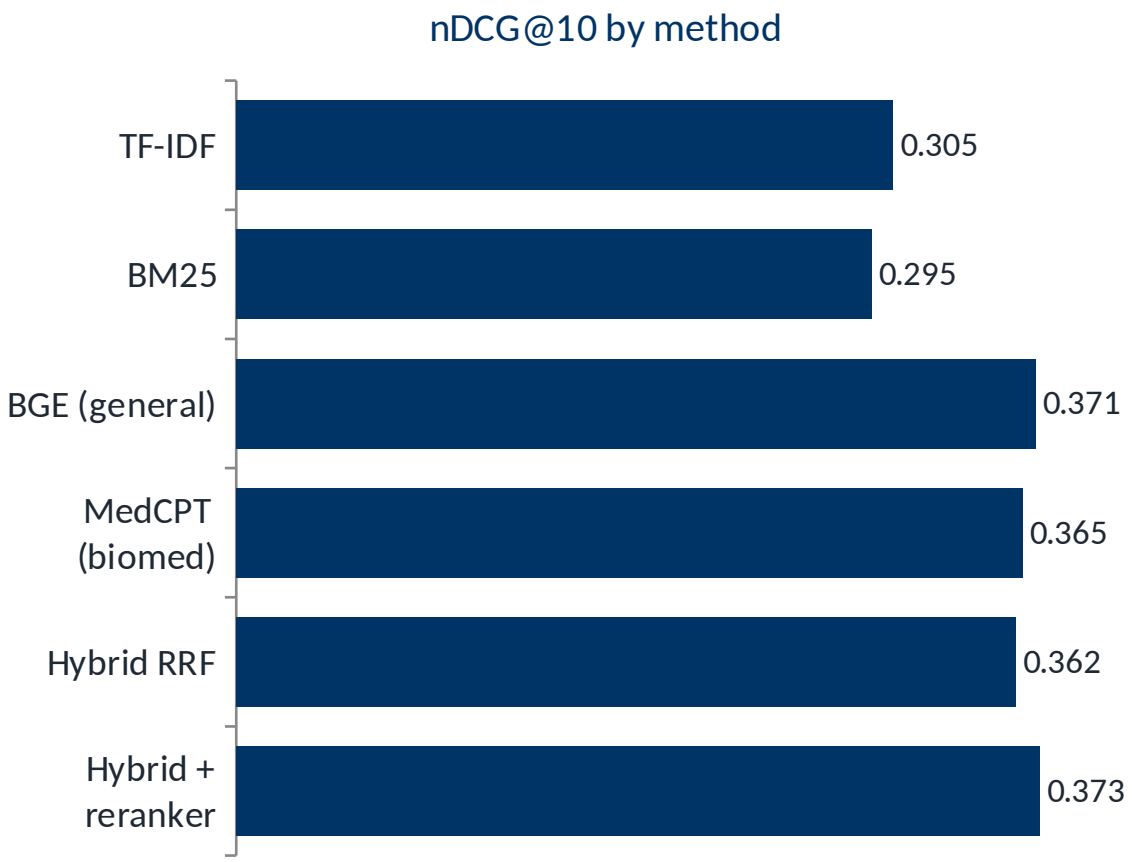


Reads query and article together. Much more accurate, much more expensive.

The trade-off: 50 full transformer passes per query, so it only reorders the top 50 RRF candidates — it cannot rescue documents outside that pool.

RESULTS · 323 TEST QUERIES

Dense retrieval is the big step; later stages add little



1 Dense beats lexical
Both dense models beat BM25 on all five metrics ($p < 0.005$).

2 A surprise baseline
Untuned TF-IDF beat BM25 (nDCG@10, $p = 0.03$).

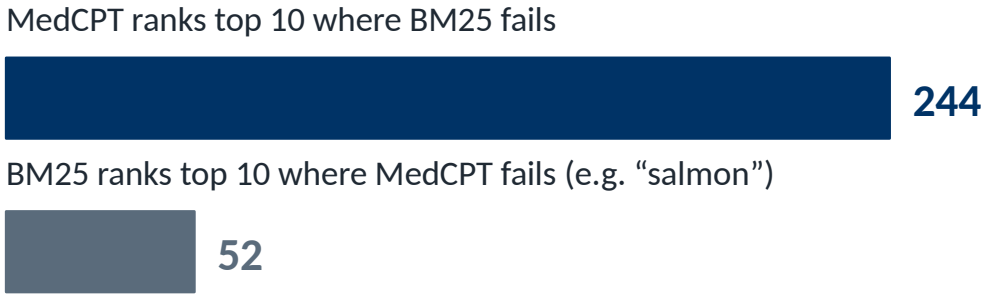
3 General ≈ biomedical
BGE and MedCPT are statistically tied (all $p \geq 0.12$).

4 Reranking is costly
Best nDCG@10 (0.373) at 1,946 ms vs 4.1 ms per query — about 480× slower.

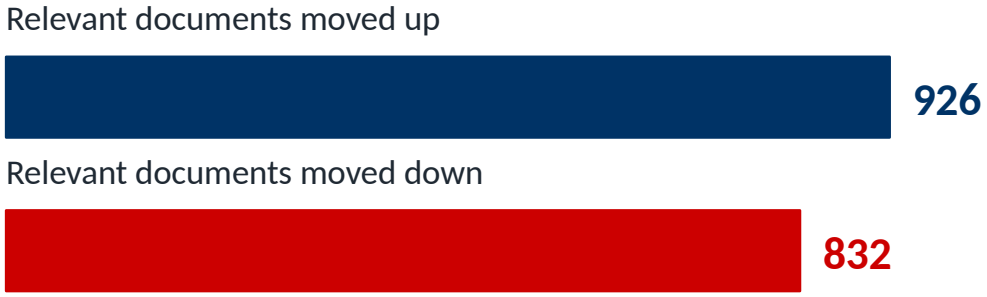
ERROR ANALYSIS & LIMITATIONS

Each method fails differently

Who rescues whom?



The reranker coin flip



Why the reranker’s nDCG@10 gain (+0.011) is not significant (p = 0.19).

Research limitations

- One collection, 323 queries — small effects are hard to detect.
- Rerank pool capped at 50 to keep latency manageable.
- BM25 and RRF left untuned (no dev-set tuning yet).
- Unjudged documents count as non-relevant.

REPRODUCIBILITY

Reproducible research artifacts



Live research platform

Run BM25, MedCPT and hybrid rankings side by side in the browser.

Scan or visit

biomedical-hybrid-ir.vercel.app



GitHub source code

Library, scripts, tests and results. Course demo on branch cap6776/course-demo.

Scan or visit

github.com/Arungharami/biomedical-hybrid-ir

TAKEAWAY

Biomedical Hybrid Information Retrieval

Meaning-based semantic retrieval closes the vocabulary gap between everyday questions and medical literature, while lexical precision stays a valuable fallback. **Fusion and reranking add small, uneven gains** — and reranking must justify its ~480× latency cost.

Live platform: biomedical-hybrid-ir.vercel.app

GitHub: github.com/Arungharami/biomedical-hybrid-ir